Paper 6

Rest, Uniform Motion, and Inertia as Active Internal Ledger States

Paper 6 of the Series

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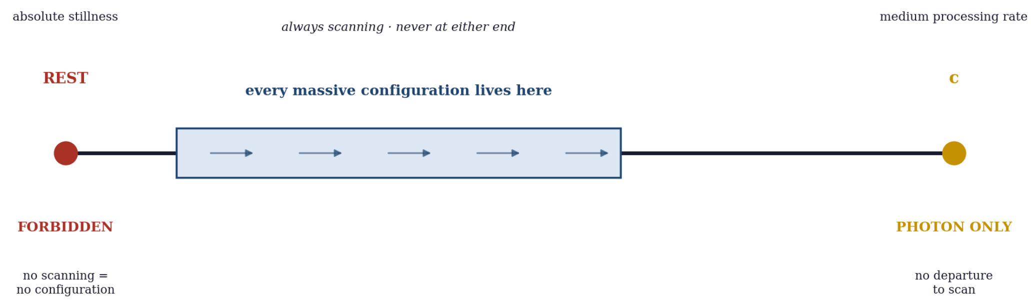
ABSTRACT

Paper 5 fixed the law of stability: a configuration is closed when its identity scan locks, and the open states endure as searches. This paper takes the next question the ruler must answer — what are rest and motion — and returns an answer that breaks with the passive picture inherited from classical mechanics. In this framework there is no rest. A configuration is a localized geometry of the continuous medium, and a geometry that has departed from the photon baseline runs its manifold scan without pause, querying its neighboring shapes on every tick of the medium's processing rate to know what it is and what it must resolve toward. A configuration that stopped scanning would stop existing as that configuration; absolute rest is not merely unobserved but structurally forbidden, in the same way that absolute zero motion is forbidden — the only thing that does not run a temporal scan is the photon, and the photon is not at rest, it is at c. What classical physics calls "rest" is therefore an active state: a configuration whose scan cycle is balanced symmetrically against the surrounding medium, holding net position while updating ceaselessly. What it calls "uniform motion" is the same active cycle carrying a directional offset — stepped along a vector across the medium, self-sustaining once set because nothing in a balanced or offset cycle demands an external push to continue. The two are distinct internal ledger conditions, not one condition in two reference frames. The paper then meets the obvious objection head-on: if the two states differ internally, why can no experiment inside a moving frame detect the difference? Because observer and apparatus are the same medium, scaled identically by their shared surroundings, so the relative ledger between them is unchanged — the framework recovers the relativity principle exactly, as a consequence of the partner-scan mechanics rather than as an axiom about spacetime. Inertia follows directly: resistance to a change of state is the structural workload of rewriting an established scan cycle, and it is named here and handed to Paper 7 for the accounting of acceleration and momentum.

1. Purpose: There Is No Rest

Figure 1. The Limit Pair: Rest and c Are Both Forbidden to Mass



One prohibition, two ends: a massive thing can reach neither rest nor c.

Figure 1. The prohibition of rest as the lower partner of a limit pair: a massive configuration can reach neither absolute rest (no scanning, hence no configuration) nor c (no departure, hence the photon). It lives perpetually between the two ends it cannot touch.

The classical picture of motion rests on a state the framework does not permit: a body at rest, inert and unchanging until a force acts on it, requiring no activity to remain as it is. Everything in this series denies that such a state can exist. A configuration is not an object sitting in space; it is a localized geometry of the continuous medium, and geometry that has departed from the photon baseline is defined by the scan it runs (Papers 4, 5). A configuration that ran no scan would have no identity to hold and no resolution to seek — it would not be a quiet version of itself, it would not be itself at all.

So the premise of this paper is stated plainly at the outset, because everything follows from it: nothing that carries mass can be at absolute rest, just as nothing that carries mass can reach c. The two prohibitions are the same prohibition read at its two ends. Only the photon escapes both — it runs no temporal scan and it travels at c — and the photon is precisely not a counterexample, because the photon is the baseline, the one state that is not a departure. Every departed configuration, every massive thing, is caught between the two limits it can never reach: it can be neither perfectly still nor at the speed of the medium's own processing. It is always active, always scanning, always querying its neighbors. Rest is not rare or hard to achieve; it is forbidden by the hardware.

There is a second, sharper reason a departed configuration can never fall silent, and it is worth stating now because it underwrites everything that follows. A configuration that has left the baseline must run its manifold scan ceaselessly for two inseparable reasons of pure accounting: to know, on every tick, that it is *not* the photon — that it still holds a departure to account for — and to know how that departure could be resolved back toward baseline. Both are knowledge the configuration can hold only by continuously scanning. The moment it stopped, it would have no register of its own departure and no map back to the zero; it would not become a still version of itself, it would lose the very accounting that constitutes it as a departed thing. The scan is not a behavior the configuration performs in addition to existing — it is the running ledger by which the configuration knows it is something other than baseline and knows what return would cost. Stop the ledger and there is no departure being tracked, which is to say there is no configuration, only baseline. Rest is forbidden because a departure that is not being accounted for is not a departure at all.

The contract of this paper. This paper *establishes* that rest and uniform motion are two distinct active states of one ceaseless internal ledger, and that the relativity principle follows as a consequence of the partner-scan mechanics rather than standing as an independent axiom. It *assumes* from Papers 4 and 5 the two scans and their operational definitions, the photon baseline, the Closure Principle, mass as displacement from the sovereign zero, and energy as added geometry conserved through relocation. It does *not* calculate the magnitude of inertia or the momentum of a moving configuration — that accounting is Paper 7 — nor the dilation of the scan cycle into a quantitative time law, which is Paper 13, nor gravity, which is Paper 10. What it provides is the mechanical account of what rest and motion *are*, on which those papers compute.

2. The Partner-Scan Architecture

The reason rest is forbidden is the reason configurations must scan at all, and that reason is the continuity of the medium. The fabric is not a vacuum holding isolated objects; it is a dense, continuous arrangement of geometric shapes, and a resolved configuration is a feature of that arrangement, not a tenant within it. A feature of a continuous medium cannot be isolated from its surroundings, because it is made of the same fabric as its surroundings and meets them at every boundary.

This is why the scans of Papers 4 and 5 are not solitary loops a configuration runs against itself. They are *partner queries*. The identity scan verifies what the configuration is, and the temporal scan tracks what it must resolve toward, but neither can be answered in isolation: to know what it is, a configuration must read what surrounds it, and to know what it must resolve toward, it must read what is available to resolve with. On every tick of the medium's processing rate — the rate whose visible face is c (Paper 4) — the twin scans query the immediately neighboring shapes: what is in proximity, and in what state. The continuity of the fabric makes this query unceasing, because there is no tick on which a feature of a continuous medium has no neighbors to read.

A configuration is therefore a perpetual active processing event, not a static thing that happens to be monitored. The querying is not maintenance performed upon an object that would otherwise sit still; the querying is what the configuration *is*. The word *query* must be stripped of any suggestion of intent, because the mechanism is not a choice the configuration makes. A partner query is not an act of asking; it is the local, automatic resolution of a geometric gradient across a continuous boundary — the same way any continuous medium settles a difference between adjacent regions without anything deciding to settle it. The scan is the inevitable throughput of a continuous fabric maintaining its own topological continuity: where there is a boundary between a configuration and its surroundings, the medium reconciles across that boundary every tick, because a continuous medium cannot hold an unreconciled discontinuity in place any more than a taut surface can hold a crease that nothing pins. Nothing in the configuration wants to scan, and nothing decides to. The scan is a boundary condition of continuity itself, and the prohibition of rest follows from it not as a rule imposed on the configuration but as a fact about what a continuous medium mechanically is. This is the mechanical root of the prohibition of rest: the scans never stop, because a feature defined by its relationship to a continuous surrounding medium is never not in that relationship. To stop scanning would be to stop being a feature of the medium — to resolve away entirely — not to come to rest. Rest, as a state of existing without scanning, is not on the menu of things the hardware can do.

There is a sharper way to see why the scan can never pause, and it turns on what the scan is *for*. A departed configuration must run its manifold scan ceaselessly to answer two questions it can never stop needing answered. The first is *what it is*: specifically, that it is not the photon. A configuration that is a standing departure from baseline holds an identity only by continuously confirming that departure against its surroundings — the moment it stopped confirming, there would be nothing distinguishing it from the baseline it departed from, and it would have no held identity at all. The second is *how to return*: the temporal scan is the search for the conserving partners through which the configuration could resolve back toward baseline, and that search cannot be suspended, because a configuration with no live accounting of its route to resolution is a configuration with an unbalanced ledger and no way to balance it. To be a massive thing is to be perpetually answering both questions — confirming it is not the photon, and searching for the path back to the photon — and absolute rest would require abandoning both. A configuration cannot stop asking what it is and how to resolve and still be a configuration. That is why the prohibition is absolute and not merely practical: the scan is identity and resolution themselves, and neither can be paused without ceasing to exist.

This also sharpens the partner principle that Paper 5 used without naming: resolution always requires partners because *identity itself* requires partners. A configuration does not first exist and then look for something to resolve with; it holds its identity only by continuous reference to what surrounds it. The search Paper 5 described — the open state sweeping for a conserving partner — is the same partner-query running with a deficit to fill. Closed and open configurations alike query without pause; the closed one reads a settled relationship, the open one reads a reaching one, but neither is ever silent.

3. The Two Active States

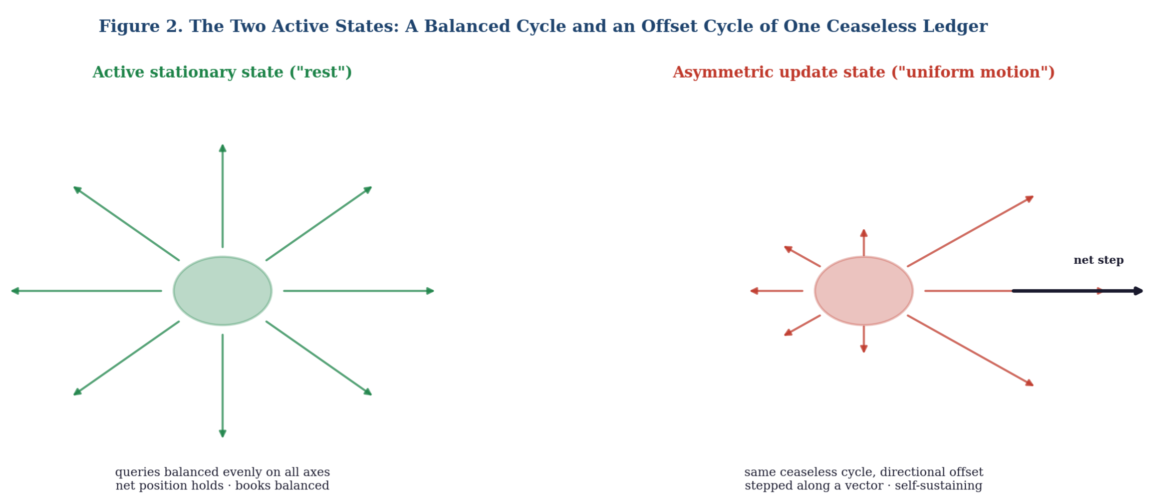


Figure 2. The two active states of one ceaseless ledger: the balanced cycle queries evenly on all axes and holds net position; the offset cycle carries a directional bias stepped along a vector, self-sustaining once set.

Because rest is banned, the two conditions classical physics conflated under "states of motion" must be redescribed as what they actually are: two cycles of the same ceaseless ledger, distinguished by their symmetry.

The active stationary state — what classical physics calls rest. Here the configuration's twin scans query their neighboring shapes in a balanced, symmetrical cycle: the coordinate updates register evenly across the three axes, with no net directional bias from one tick to the next. The configuration is doing an enormous amount of work — every tick is a full partner query and a full internal verification — but the work is balanced, so its net position relative to the surrounding medium does not change. To a macroscopic observer the configuration looks static, and the illusion is total, because the only thing an outside observer can see is the net position, which holds. What the observer cannot see is the intense, high-frequency coordinate conversation the configuration is conducting with its proximity on every tick to *hold* that position. Stillness is the appearance of a balanced books, not the absence of transactions.

The asymmetric update state — what classical physics calls uniform motion. Here the same ceaseless cycle carries a directional offset. As the twin scans query their partners, the sequence of updates is stepped along a vector: each tick, the configuration's relationship to its neighbors is rewritten slightly forward in one direction, so that across many ticks the configuration's net position translates across the medium. The configuration is not sliding through an empty void, because there is no void; it is rewriting its identity from one neighboring shape to the next, continuously, the way a pattern moves across a medium by being reconstituted in successive positions rather than by a substance traveling. And it maintains its rate without any external force, because once the cycle carries its directional offset there is nothing in a continuous, conserved ledger to remove the offset: the asymmetric cycle is as self-sustaining as the balanced one. Uniform motion needs no push to continue for exactly the reason rest needs no push to continue — both are active cycles that conserve themselves, and neither is privileged as the "natural" state.

The self-sustaining character of the offset is not an assertion but a conservation law, and it answers the reviewer who expects motion to dissipate the way a velocity gradient diffuses in a classical fluid. That expectation misidentifies the medium as a sloppy continuous fluid rather than a discrete structural engine. A mass configuration is not a foreign object wading through a separate fluid; it is a bound structural knot composed of the medium itself, and the directional offset cut into its ledger is a quantized geometric deficit, not a loose substance that can leak. For the asymmetry to diffuse into the surroundings, the neighboring shapes would have to accept fractional parts of the configuration's internal value balance — but the medium outside the configuration runs pure baseline, carrying only O values, with no hardware capacity to absorb or record a partial vector transformation without violating its own conservation. The offset is a complete, indivisible ledger entry: it can be transferred in full to another closed configuration in a collision, but it has no valid path to dissolve into a zero-loaded baseline that lacks the slots to count it. This is Newton's first law derived as a conservation mandate rather than accepted as an unexplained freebie — motion does not dissipate because a quantized geometric deficit cannot bleed into a medium that cannot record it. The standard account asserts that an object in motion stays in motion and offers no internal mechanism for why; the framework supplies the mechanism, and it is the discreteness of the ledger itself.

The decisive claim of this paper is that these two are genuinely *distinct internal conditions*, not one condition described from two reference frames. The balanced cycle and the offset cycle are different geometric facts about how a configuration is querying its medium, and the difference is real at the level of the hardware. This is where the framework parts with the classical and relativistic picture, which holds rest and uniform motion to be the same state distinguished only by choice of frame. The framework says: the medium itself registers the difference, because the offset is a real directional asymmetry in the ledger, even though — as the next section establishes — no observer made of the same medium can detect it from inside.

3.1 The Illusion of the Ether Wind and the Null Result of 1887

To a reviewer accustomed to the historical aether-drift debates, the claim that the medium registers a localized ledger offset immediately evokes the ghost of a preferred reference frame — and the apparent refutation of 1887. This is a misreading of the architecture, and the structural audit of the Michelson–Morley experiment is the proof. In the nineteenth-century picture, the aether was an independent fluid container, matter was a distinct substance moving through it, and light was an external wave propagating in the background; an apparatus moving through the container at velocity v should therefore register an "aether wind," shifting the light's speed relative to the arms of the machine.

The framework inverts that metaphysics completely. As established in Paper 4, a photon is not a disembodied entity propagating through an independent space; the photon is the un-tilted baseline behavior of the continuous medium itself — the sovereign zero. The apparatus — mirrors, beam splitter, detector, the experimenter — is built of stable closed configurations made of that same fabric. When the laboratory is placed in uniform motion, every configuration in it carries the identical directional offset in its ledger. So when the experimenter sends a photon down an interferometer arm, they are not passing an independent wave through a separate fluid; they are querying the medium's own baseline with an apparatus made of the medium that is itself running the offset cycle. The measuring instruments hold their identity only by running their partner-scans against the medium's baseline, so the apparatus is structurally forced to measure the medium against itself. The offset cancels perfectly in internal comparison because the observer and the baseline ruler share the identical underlying fabric.

The framework does not dodge the null result; it hardcodes it as a geometric necessity. An internal observer can never detect an aether wind when the measuring instruments are literally constructed out of the very baseline they are trying to measure. And the inversion carries a counter-charge. Having discarded the medium on the strength of that null result, the standard account was left to clamp the invariance of c by decree — an unexplained postulate — and to warp space and time elastically to honor it. The framework does not mandate the invariance of c; it derives it as a property of the hardware, because the baseline is the medium's own processing rate and the apparatus measuring it is a knot in that same medium. The experiment had to return null, not because there is no medium, but because there is nothing else for an internal instrument to measure against.

4. Recovering the Relativity Principle

Figure 3. Why the Offset Self-Conceals: A Shared Term Cancels From Every Internal Comparison

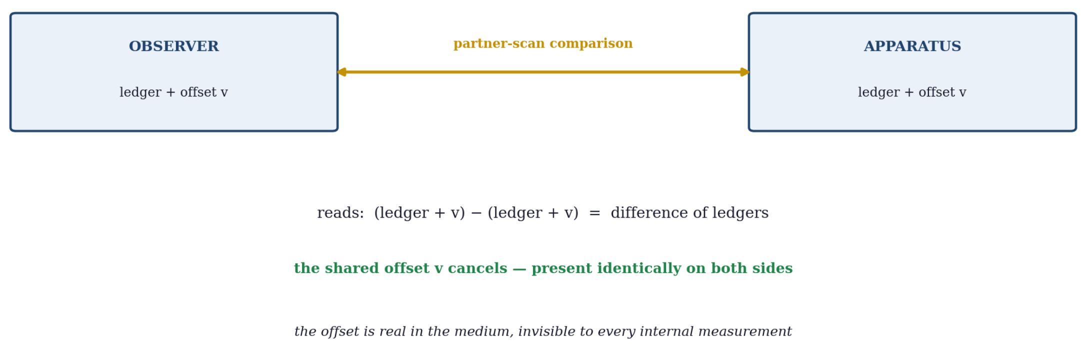


Figure 3. The relativity principle recovered as cancellation: observer and apparatus share the same offset, so a partner-scan comparison differences it away — the offset is real in the medium and invisible to every internal measurement.

The objection arrives immediately, and it is the one that would kill the paper if it had no answer: if the active stationary state and the asymmetric update state are genuinely different internal conditions, why can no experiment performed inside a uniformly moving frame detect that frame's motion? The relativity principle — that the laws of physics are identical in all uniformly moving frames, and that no internal experiment reveals uniform motion — is among the most thoroughly confirmed statements in physics. A framework that contradicted it would be wrong by breakfast.

The framework does not contradict it. It derives it. The reason no internal experiment detects the asymmetry is that the experiment, the apparatus, and the observer are all features of the same continuous medium, and all of them carry the same directional offset in their ledgers. When the observer's scans query the apparatus's scans — when one configuration in the moving frame reads another — the offset is present identically on both sides of the query, and a quantity present identically on both sides of a comparison cancels out of the comparison. The relative ledger between observer and apparatus is exactly what it would be in the balanced state, because the offset they share is invisible to a measurement made entirely between configurations that share it. Every clock in the moving frame runs on the same offset cycle; every ruler is built from configurations carrying the same offset; every process the observer could use to detect motion is itself conducted in the offset ledger. The asymmetry is real, and it is perfectly self-concealing from the inside, because there is no internal reference that does not also carry it.

This concealment holds on every axis, not only along the line of motion, and the reason must be stated because a careful experimentalist will probe exactly there. The objection is that if the offset is a vector along, say, the x-axis, a measurement taken along the orthogonal y-axis should reveal an anisotropy — the moving frame should look different across the direction of motion than along it, and isotropy tests of great precision would catch it. The objection assumes the offset is a vector tacked onto a Cartesian coordinate, affecting one axis and leaving the others untouched. It is not. The scan of Papers 4 and 5 is a *manifold* scan: it updates across the three-axis framework as one integrated unit, and the offset does not displace a single axis but alters the fundamental resolution scale at which the whole three-axis ledger is run. Because the alteration is to the integrated scale rather than to one direction, it enters every axis of the configuration's internal accounting identically — and because the measuring tool shares the alteration with the thing measured, the shared change of scale applies uniformly to both across all three axes at once. A uniform scaling shared by ruler and object cancels in every direction, not just the one the motion points along: there is no axis on which the ruler is unscaled while the object is scaled, so there is no axis on which an anisotropy could appear. The isotropy the experiments confirm is not a coincidence the framework must survive; it is what a shared manifold-scale alteration necessarily produces.

This is the relativity principle recovered exactly — and recovered as a *consequence* rather than assumed as a *postulate*. Einstein took the principle as an axiom and built spacetime to honor it. The framework takes the partner-scan mechanics as the foundation and finds the principle falling out of them: of course no internal experiment detects uniform motion, because detection is a comparison and the thing to be detected is shared by both terms of every available comparison. What the framework strips away is the spacetime apparatus erected to explain the principle. There is no warping manifold, no geometry of events, no need for the principle to be fundamental. There is a continuous medium, configurations querying their neighbors, and a directional offset that cancels out of every internal measurement while remaining a real fact about the medium. The experimental content of relativity is kept whole; the ontology beneath it is replaced.

One consequence is worth marking now and paying off later. The asymmetry that hides from internal comparison does *not* hide from the medium itself, and a directional cycle does not complete its internal identity check on the same footing as a balanced one — the offset means each verification is conducted across a stepped sequence rather than a symmetric one. That difference is the structural seed of time dilation: a moving configuration's scan cycle does not keep pace with a stationary one in the medium's own accounting, even though neither can detect the discrepancy from inside its own frame. The quantitative law of that difference is the work of Paper 13; this paper marks only that the seed is the directional offset, not a warping of time itself.

This also forecloses the charge that an offset no internal experiment can detect is an unfalsifiable hidden variable. The cancellation of Section 4 is an exact property of *uniform, unaccelerated* translation only; the offset is never permanently hidden. It surfaces through two later, quantifiable pathways. The first is time dilation (Paper 13): a configuration has a finite processing budget per tick, and a ledger carrying a directional offset spends a fixed share of that budget on the relocation workload, leaving less for the temporal scan, so the moving configuration's clock physically runs slow — the offset's direct footprint, not an effect of perspective. The second is behavior under acceleration (Paper 7): while a uniform offset self-conceals within its own co-moving baseline, the structural workload required to *change* that offset is an absolute function of the established ledger state, and it surfaces the instant the system departs from unaccelerated motion. The offset is therefore falsifiable in principle — just not by a within-frame uniform-motion experiment, which is exactly the one experiment its self-concealment forbids. A critic who calls it unfalsifiable must also call the rate of time itself unfalsifiable, since the framework binds the two.

5. Inertia as the Workload of Changing the Cycle

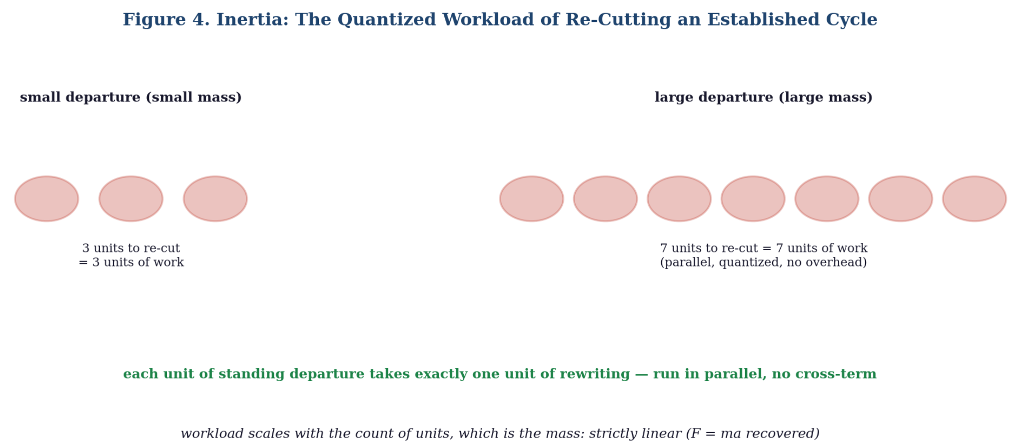


Figure 4. Inertia as quantized workload: each unit of standing departure takes exactly one unit of rewriting, run in parallel with no cross-term, so the cost of changing motion scales linearly with the count of units — that is, with mass.

With rest and uniform motion redescribed as a balanced cycle and an offset cycle, inertia is no longer a mysterious tendency of matter to resist acceleration. It is the structural cost of rewriting an established ledger.

A configuration in a balanced cycle, or in an offset cycle, is running a self-sustaining pattern of partner queries that conserves itself tick after tick. To change that state — to start a stationary configuration moving, to speed or slow or turn a moving one — is to impose a *different* offset on the cycle, and that is not free: the existing pattern of relationships with the surrounding medium must be rewritten into a new pattern, and rewriting geometry is added geometry, which is energy (Paper 4). The resistance a configuration presents to a change of motion is the workload of that rewriting — the amount of ledger that must be re-cut to move the configuration from its current cycle to the demanded one.

The conversion from an external push to a discrete ledger step is itself mechanical, not a magic word. When external geometry is delivered into a configuration, its boundary is forced into a calculation mismatch — the incoming displacement cannot be reconciled with the existing balanced cycle while the configuration maintains its closure (Paper 5). To stay closed, the configuration must clear the mismatch the only way conservation allows: by writing a permanent directional offset step into its update ledger, absorbing the delivered geometry as a change of cycle. Work is that forced rewrite. This is why work converts cleanly into a discrete offset rather than smearing away as a continuous loss: the configuration cannot hold the delivered geometry as anything other than a re-cut of its own cycle, because it has no inert storage and no option but closure.

This is why inertia is proportional to mass, without any separate postulate connecting them. Mass is uncompensated displacement from the sovereign zero (Papers 4, 5) — the size of the configuration's standing departure, the quantity of geometry it holds away from baseline. A larger departure is a larger established ledger, and a larger ledger costs more to rewrite. The same quantity that measures how far the configuration sits from baseline measures how much cycle must be re-cut to change its motion, so inertial resistance and mass are not two properties that happen to be equal — they are one quantity read in two situations, exactly as Paper 4.5 argued inertial and gravitational mass are one quantity rather than two clamped equal.

The proportionality is strictly linear, and the reason is structural, not assumed. A critic will ask why re-cutting a larger ledger does not incur exponential overhead or internal self-interference — why doubling the standing departure does not more-than-double the workload. It does not, because the ledger updates are quantized and additive, and they run in parallel. A configuration's standing departure is built of discrete, independent partner-query units (Papers 2, 3), and rewriting the cycle re-cuts each unit separately: one unit of standing departure takes exactly one unit of geometric rewriting, with no cross-term between units, because each unit's re-cut is a local operation completed against its own neighbors on the same tick. There is no computational drag accumulating across the configuration and no self-interference between units, because the medium's hardware runs all the local re-cuts simultaneously rather than in sequence — the workload scales with the *count* of units, which is the mass, and with nothing else. This is why the relationship is linear and not merely monotonic: the cost of changing motion is a sum of identical independent quantized costs, one per unit of departure, which is the exact structural meaning of proportionality to mass. The configuration resists changing its motion for the same reason it has mass at all: both are the standing cost of its departure from the photon.

The quantitative accounting — how much workload for how much change, the momentum a moving configuration carries, the relation between applied interaction and resulting change of cycle — is the subject of Paper 7, and this paper does not pre-empt it. What this paper fixes is the mechanical identity of inertia: it is not a property matter has, it is the cost of rewriting what matter is doing.

6. What This Paper Establishes, Pointing Forward

The ruler now has its account of motion, and it is an account with no passive states in it. Rest is an active balanced cycle; uniform motion is an active offset cycle; both run ceaselessly because a feature of a continuous medium can never stop querying its neighbors without ceasing to exist. The relativity principle is recovered as the self-concealment of a shared offset, not assumed as an axiom about spacetime. Inertia is the workload of rewriting a cycle, one quantity with mass rather than a separate property.

Each thread is handed forward to the paper that computes it. Paper 7 takes inertia and the offset cycle and builds the accounting of acceleration and momentum — the quantitative law of what it costs to change a cycle and what a changed cycle carries. Paper 8 takes mass as displacement and computes the internal accounting of departure from c. Paper 10 takes the partner-query and the resolution-seeking of the open states and develops gravity as the macroscopic face of that seeking. Paper 13 takes the directional-offset seed marked in Section 4 and builds the law of time dilation as a property of the scan cycle, not of a warping manifold. Each inherits the same premise this paper fixed: there is no rest, only the ceaseless active ledger, balanced or offset, querying its partners on every tick of the medium's clock.

7. Conclusion

Classical mechanics began with a body at rest and asked what it takes to move it. This paper begins by denying that a body at rest exists. A configuration is a localized geometry of a continuous medium, and it holds its identity only by querying its neighbors without pause; the scan never stops, so rest — existence without scanning — is forbidden by the same hardware that forbids a massive configuration from reaching c. What classical physics called rest is an active stationary state, a cycle balanced symmetrically against the medium so that net position holds while the books turn over ceaselessly. What it called uniform motion is the same cycle carrying a directional offset, self-sustaining because a conserved ledger does not shed its own asymmetry. The two are distinct conditions of the medium, and the medium registers the difference — yet no observer inside a moving frame can detect it, because the offset is shared by every configuration the observer could measure against, and a shared quantity cancels from every internal comparison. So the relativity principle stands, recovered exactly, stripped of the spacetime apparatus and grounded in partner-scan mechanics. And inertia, in this account, is not a property of matter but the cost of rewriting matter's cycle — the same standing departure that gives a configuration mass, read as the workload of changing what that configuration is doing. There is no rest. There is only the ledger, and it never stops.

References

This paper is a Tier 1 foundational paper of the *Departure From the Photon Baseline* series and builds on the framework established in the companion papers below.

1. Kadunic, D. (2026). *The 27 Lagrangians of Gravity: A Survey of the Field's Standards of Evaluation* (Paper 1). Zenodo. https://doi.org/10.5281/zenodo.20434286
2. Kadunic, D. (2026). *The Three-Axis Framework and the Nine-State Classification of the 27 Geometric States* (Paper 2). Zenodo. https://doi.org/10.5281/zenodo.20510019
3. Kadunic, D. (2026). *They Assumed It; They Did Not See It: A Critique of Inferred Particles in Standard Physics* (Paper 2.5). Zenodo. https://doi.org/10.5281/zenodo.20518514
4. Kadunic, D. (2026). *The Nine-State Classification with Inside-Outside Structure* (Paper 3). Zenodo. https://doi.org/10.5281/zenodo.20561941
5. Kadunic, D. (2026). *The Constant That Does Not Stay Constant: A Structural Audit of the Fine-Structure Measurement* (Paper 3.5). Zenodo. https://doi.org/10.5281/zenodo.20575611
6. Kadunic, D. (2026). *The Photon as Baseline, Electron-Positron as First Tilt* (Paper 4). Zenodo. https://doi.org/10.5281/zenodo.20633821
7. Kadunic, D. (2026). *The Fragmented Metric — A Structural Deconstruction of Targeted Ruling and the Proliferation of Unfalsifiable Ontologies* (Paper 4.5). Zenodo. https://doi.org/10.5281/zenodo.20634735
8. Kadunic, D. (2026). *The Closure Principle and Configuration Stability* (Paper 5). Zenodo. https://doi.org/10.5281/zenodo.20690435

Paper 4 establishes the photon baseline, the two scans and their operational definitions, mass as displacement from the sovereign zero, energy as added geometry, and propagation at c as the medium's processing rate. Paper 5 establishes the Closure Principle, the holding-versus-stability distinction, and the partner requirement for resolution that this paper grounds in the partner-scan architecture. Paper 4.5 establishes the single-ledger reading of inertial and gravitational mass that Section 5 draws on. The quantitative accounting of acceleration and momentum is the work of Paper 7; mass as departure from c is Paper 8; gravity as resolution-seeking is Paper 10; the law of time dilation is Paper 13; none is pre-empted here.

A. The Prohibition of Rest, Stated as a Limit Pair

This appendix states the central prohibition of the paper as a matched pair of limits, to show that the impossibility of rest is not an independent assertion but the lower partner of a prohibition the series already carries at its upper end.

The framework forbids a massive configuration from reaching two conditions, and the two prohibitions have one mechanical source.

The upper limit is c. A massive configuration cannot reach the speed of the medium's own processing rate, because to do so it would have to shed its entire standing departure and run only the identity scan — which is to say, it would have to become the photon. The speed drop off baseline established in Paper 4 is the near end of this limit: the more a configuration's energy is committed to its standing departure and its temporal-scan accounting, the further below c it sits, and the full departure can never be zero for a massive thing.

The lower limit is rest. A massive configuration cannot reach absolute stillness, because to do so it would have to stop querying its neighbors — stop running the ceaseless cycle by which it holds its identity — which is to say, it would have to cease being a configuration of the medium at all. Absolute rest is not a low-energy state a configuration could settle into; it is the absence of the scanning that constitutes the configuration.

The two limits are one prohibition. A massive configuration is a standing departure from the photon that must run its full manifold scan ceaselessly: it can be neither at the baseline's speed (c, where there is no departure to scan) nor at absolute rest (where there is no scanning to hold the departure). It is always in motion in the only sense that matters to the hardware — always scanning, always querying, always active — and "rest" and "c" are the two unreachable ends between which every massive thing perpetually runs. Only the photon occupies an end, and it occupies the upper one, at c, by being the baseline rather than a departure from it. Nothing occupies the lower end, because the lower end is non-existence wearing the mask of stillness.